



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2024-25
MONDAY TEST

CLASS : X
SUBJECT : CHEMISTRY

FULL MARKS : 40
DATE: 22.04.2024

General Instructions:

- The paper consists of three printed pages.
- Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.

SECTION A
(Attempt all questions)

Question: 1

Choose the correct answer from the options given below: [8]

- (i) Chlorine is the element below fluorine in the long form of periodic table. The electronegativity of chlorine will be:
- More than fluorine
 - Less than fluorine
 - Same as fluorine
 - None of these
- (ii) The metal hydroxide which reacts with both acids and alkalis to form salt and water is:
- Calcium hydroxide
 - Magnesium hydroxide
 - Aluminium hydroxide
 - Ferric hydroxide
- (iii) Which of the following pair of compounds have same vapour densities:
- Nitrogen dioxide and Sulphur dioxide
 - Nitrous oxide and Carbon dioxide
 - Carbon monoxide and Nitrous oxide
 - Nitric oxide and Carbon monoxide
- [C=12, N=14, O=16, S=32]
- (iv) The most electronegative element of period 3
- Argon
 - Phosphorus
 - Chlorine
 - Sulphur
- (v) In the molecule of water, oxygen atom has:
- One shared pair of electrons
 - Three shared pair of electrons
 - Two lone pair of electrons
 - One lone pair of electrons

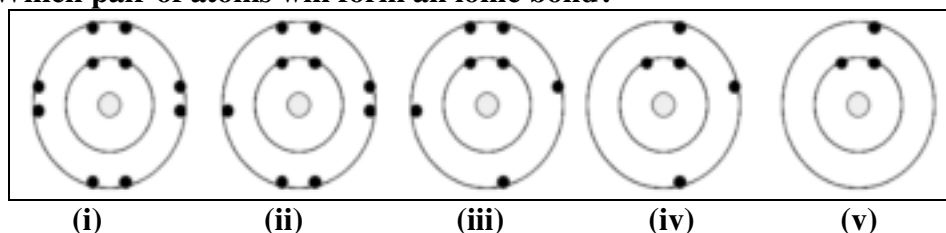
(vi) Electron affinity is maximum in:

- A. Lithium
- B. Magnesium
- C. Argon
- D. Bromine

(vii) The reagent which can be used to distinguish between lead nitrate solution and zinc nitrate solution is:

- A. Sodium hydroxide solution only
- B. Ammonium hydroxide solution only
- C. Both sodium hydroxide and ammonium hydroxide
- D. Neither of them

(viii) Which pair of atoms will form an ionic bond?



- A. (i) and (v)
- B. (iii) and (v)
- C. (ii) and (iii)
- D. (i) and (iv)

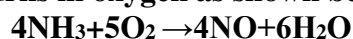
Question: 2

(i) Draw the electron dot structure of:

- (a) Carbon tetrachloride
- (b) Ammonium ion
- (c) Magnesium oxide
- (d) Nitrogen

[4]

(ii) Ammonia burns in oxygen as shown below:



If 240cc of ammonia is burnt in 300cc of oxygen, find out the composition of the resultant gaseous mixture at room temperature.

[3]

SECTION B

(Attempt all questions)

Question: 3

(i) The following table shows the electronic configuration of the atoms A, B, C & D.

Element	A	B	C	D
Electronic configuration	2,8,8,2	2,6	2,8,7	2,4

(a) Write the formula of the compound between:

- 1. A and B
- 2. D and C

(b) Which of them is an alkaline earth metal?

- (ii) Arrange the following as per the instructions given in the bracket:
(a) Carbon, Fluorine, Beryllium (decreasing order of atomic size)
(b) Potassium, Lithium, Sodium (increasing order of ionisation potential) [5]

Question: 4

- (i) How would you distinguish between the following pairs of compounds using a suitable reagent:
(a) Zinc nitrate solution and lead nitrate solution
(b) Ferrous sulphate solution and ferric chloride solution.
(c) Calcium chloride solution and magnesium chloride solution
- (ii) Calculate the percentage of water of crystallisation in $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$.
[Mg=24, S=32, O=16, H=1] [5]

Question: 5

- (i) A compound of carbon, hydrogen and oxygen is found to contain 40% of carbon, 6.7% hydrogen and 53.3% of oxygen. Calculate the molecular formula of the compound if its vapour density is 30. [C=12, O=16, H=1]
- (ii) Give reasons for the following statements
(a) Inert gases do not form ions.
(b) Covalent compounds have a low melting and boiling point. [5]

Question: 6

- (i) In Period 3 of the periodic table element A is placed to the left of element B. On the basis of the information, choose the correct word from the brackets to complete the following statements :
(a) The element B would have (lower/higher) metallic character than A.
(b) The element A would probably have (lesser/higher) electron affinity than B
(c) The element A would have (greater/smaller) atomic size than B.
- (ii) Define the following terms:
(a) Gay Lussac's Law
(b) Electronegativity [5]

Question: 7

- (i) Complete the following statements by choosing a suitable word from the bracket.
(a) Water is a _____ (polar/ non polar) covalent molecule.
(b) Lithium is an _____ (alkali/ alkaline earth) metal.
(c) Calcium oxide is a _____ (ionic/ covalent) molecule.
- (ii) Students tried to burn 600 cc of ethane in 2500cc of oxygen in the laboratory. According to them how much will be the volume of unused oxygen and volume of carbon dioxide formed. The equation for the reaction is:
$$2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$$
 [5]